

Signature:

I hereby confirm that all the pages of my exam paper
have been stapled to this cover sheet.

End-of-Module Exam: 2026 Autumn Semester

Zürcher Hochschule

für Angewandte Wissenschaften

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Number of supplementary answer sheets attached:

Total number of added pages:

Module:	w.BA.XX08-WIN.XX / Data Science Introduction
Exam date / time:	20.01.2027 / 14:00 – 15:00
Duration:	60 Minutes
Total number of pages:	2
Permissible resources:	Closed book
Calculator:	no calculator
Dictionary:	Yes

General information:

- Use a black or blue pen (invisible ink is not allowed) to write your answers.
- All options of a question (i.e. A–D) can be ticked.
- ONLY tick ONE option per question in the table.
- Scoring: each correctly answered question = 1 point.
- Write your answers only on the answer sheet provided; rough work on question pages is not scored.
- Switch off your mobile phone and put it in your bag for the duration of the examination.
- If you want to write a draft, use the back of the question pages. Only the answer sheet counts.
- Please remain seated until all exams have been collected.

Marked by:	Total points:	Grade:

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Grading Table

Performance Assessment	Weighting	Grade	Weighted Grade
Performance Assessment 1	100%		
		<i>(Grade on cover sheet)</i>	
Module grade (Evento)			⇐ Rounded ⇐

Marking Table

No.	Task	Points	Corrector's, initials
1	MC written Exam	30 Points	hitz
	Total	30 Points	

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Answer sheet

ONLY the answer sheet will be scored. All options of a question (i.e. A–D) can be ticked.

ONLY tick ONE option per question in this table.

Scoring: Each correctly answered question is worth **1 point**.

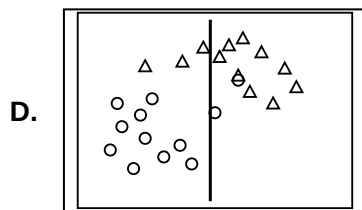
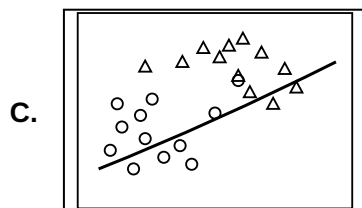
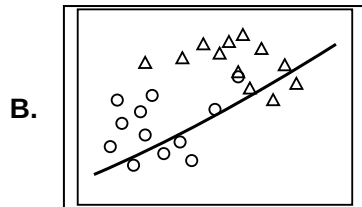
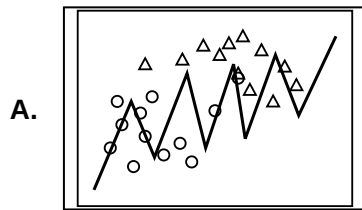
Example: For question 0, C should be ticked, then it would look like this.

Question	Answer	Question	Answer
1.	☐ A ☐ B ☐ C ☐ D	16.	☐ A ☐ B ☐ C ☐ D
2.	☐ A ☐ B ☐ C ☐ D	17.	☐ A ☐ B ☐ C ☐ D
3.	☐ A ☐ B ☐ C ☐ D	18.	☐ A ☐ B ☐ C ☐ D
4.	☐ A ☐ B ☐ C ☐ D	19.	☐ A ☐ B ☐ C ☐ D
5.	☐ A ☐ B ☐ C ☐ D	20.	☐ A ☐ B ☐ C ☐ D
6.	☐ A ☐ B ☐ C ☐ D	21.	☐ A ☐ B ☐ C ☐ D
7.	☐ A ☐ B ☐ C ☐ D	22.	☐ A ☐ B ☐ C ☐ D
8.	☐ A ☐ B ☐ C ☐ D	23.	☐ A ☐ B ☐ C ☐ D
9.	☐ A ☐ B ☐ C ☐ D	24.	☐ A ☐ B ☐ C ☐ D
10.	☐ A ☐ B ☐ C ☐ D	25.	☐ A ☐ B ☐ C ☐ D
11.	☐ A ☐ B ☐ C ☐ D	26.	☐ A ☐ B ☐ C ☐ D
12.	☐ A ☐ B ☐ C ☐ D	27.	☐ A ☐ B ☐ C ☐ D
13.	☐ A ☐ B ☐ C ☐ D	28.	☐ A ☐ B ☐ C ☐ D
14.	☐ A ☐ B ☐ C ☐ D	29.	☐ A ☐ B ☐ C ☐ D
15.	☐ A ☐ B ☐ C ☐ D	30.	☐ A ☐ B ☐ C ☐ D

MC Data Science Introduction (30 Points)

1. Which single statement about VSM and System Dynamics is correct? (1)
 - A. VSM and System Dynamics are identical approaches.
 - B. The VSM is a recursive structure for viability, while SD simulates behaviour over time.
 - C. Both reject the concept of feedback.
 - D. System Dynamics is a static organisational chart.
2. Which statement about Ashby adaptation is NOT correct? (1)
 - A. Failure to adapt can let the environment dominate the system.
 - B. The only valid response is to reduce one's own variety.
 - C. Adaptation aligns internal variety with environmental variety.
 - D. A system can amplify its own variety in order to adapt.
3. Which statement about distinguishing Data Science from related terms is NOT correct? (1)
 - A. Analytics is one component within the broader field of Data Science.
 - B. Big Data and Data Science are identical concepts.
 - C. Big Data concerns the four Vs: Volume, Variety, Velocity and Veracity.
 - D. Data Science is transdisciplinary (methods, theory and application).
4. Which of the following best describes a real-time adaptive model? (1)
 - A. A model trained once and never updated again.
 - B. A static end-of-month report.
 - C. A spreadsheet pivot table.
 - D. A model that continuously adapts to streaming data via feedback.

5. Which of the following illustrations shows a classification model that suffers most from overfitting? (one answer) (1)



6. Which single statement about model-output visualization is correct? (1)
- A. Visualizations are immune to the designer's bias.
 - B. Visual literacy is never needed to interpret a chart.
 - C. Poorly designed visuals can distort interpretation and imply false completeness.
 - D. Charts can never be used to mislead.
7. Which statement about complexity is NOT correct? (1)
- A. Complex systems have many interdependent variables.
 - B. Complexity is the same as a complicated machine with many parts.
 - C. Complexity affects the value and interpretability of data.
 - D. CPS data is often non-IID (dependent across time and context).
8. Which statement distinguishing class weighting from sampling is NOT correct? (1)
- A. Oversampling duplicates minority-class examples.
 - B. SMOTE interpolates new synthetic minority samples.
 - C. Class weighting duplicates minority-class examples.
 - D. Undersampling shrinks the majority class.

9. A team says 'we have lots of data - what can we build?'. A target-oriented coach would reframe this as... (1)
- A. start from the goal and design data, features and metrics around it.
 - B. choose the most complex algorithm available.
 - C. skip stakeholder alignment to save time.
 - D. keep exploring the data until something emerges.
10. A clinician needs to know WHY a model flagged a patient as high-risk. What does this primarily require? (1)
- A. A reinforcement signal.
 - B. More unlabeled data.
 - C. Explainable AI / interpretability.
 - D. A black-box model.
11. Which single statement about CPS control under Ashby's Law is correct? (1)
- A. A controller with less variety than the system can still fully control it.
 - B. Control needs no data or feedback.
 - C. Model-based control needs no models.
 - D. CPS control is rooted in cybernetics, and the controller must match the system's variety.
12. Which statement about the VSM is NOT correct? (1)
- A. It is recursive across levels.
 - B. It uses anti-oscillatory feedback.
 - C. It supports adaptability and learning.
 - D. It is identical to a static organisational chart with no feedback.
13. Which statement about handling class imbalance is NOT correct? (1)
- A. Accuracy is the most reliable metric under strong imbalance.
 - B. Oversampling enlarges the minority class.
 - C. Undersampling shrinks the majority class.
 - D. Precision, recall and F1 are more informative than accuracy here.
14. In the VUCA framework, 'uncertainty' means... (1)
- A. many interrelated variables.
 - B. rapid, unstable change.
 - C. future outcomes are unknown despite available data.
 - D. unclear meanings with no precedents.
15. A retailer groups customers by behaviour with no predefined labels. This is... (1)
- A. reinforcement learning.
 - B. linear regression.
 - C. supervised classification.
 - D. unsupervised clustering.

16. Which single statement about requisite variety is correct? (1)
- A. A controller needs at least as much variety as the system it controls.
 - B. Variety simply equals the sheer volume of data.
 - C. More data automatically equals requisite variety.
 - D. Environmental complexity has no effect on a system's survival.
17. Which statement about System Dynamics is NOT correct? (1)
- A. It is a static, time-free reporting method.
 - B. It models behaviour over time.
 - C. Vensim/STELLA are typical modeling tools.
 - D. It uses stocks, flows and feedback loops.
18. Which statement about smart factories / Industry 4.0 is NOT correct? (1)
- A. Data Science adds anomaly detection and stream processing.
 - B. They are a key application of cyber-physical systems.
 - C. They avoid analytics entirely.
 - D. Control follows cybernetic feedback principles.
19. On a strongly imbalanced dataset, which single metric is most misleading on its own? (1)
- A. F1 score
 - B. Accuracy
 - C. Precision
 - D. Recall
20. K-medoids differs from K-means mainly because... (1)
- A. its cluster centre is an actual data object, making it more robust to outliers.
 - B. it needs labeled data.
 - C. it cannot form clusters.
 - D. it is a supervised method.
21. Which single statement about the target-centric approach is correct? (1)
- A. It recommends starting from whatever data is available and ignoring goals.
 - B. Activities, data and metrics are structured around clear business objectives.
 - C. It avoids stakeholder alignment.
 - D. It is identical to a purely system-centered approach.
22. Which statement about clustering and network analysis is NOT correct? (1)
- A. Network analysis focuses on connections and structure.
 - B. Communities in graphs resemble clusters.
 - C. Both require labeled data to function.
 - D. Clustering groups similar data points.

23. Which statement about Big Data myths is NOT correct? (1)
- A. Veracity (reliability) is an essential dimension of Big Data.
 - B. Big Data and Data Science are exactly the same thing.
 - C. 'Data lakes are inherently valuable without metadata' is a myth.
 - D. 'Merging sources automatically yields a Golden Record' is a myth.
24. Exploratory Data Analysis (EDA) is mainly about... (1)
- A. writing the business case.
 - B. tuning hyperparameters only.
 - C. deploying the final model.
 - D. understanding the data (distributions, missing values, outliers) before modeling.
25. Which is the CORRECT reading of Ashby's Law (avoiding misinterpretation)? (1)
- A. It is a precise numeric equation to be solved.
 - B. Match variety - either by amplifying your own variety or by attenuating the environment's.
 - C. Variety simply equals dataset size.
 - D. Always maximize your own complexity.
26. Which single statement about the VUCA environment is correct? (1)
- A. The four dimensions are interchangeable.
 - B. Preparation is futile under VUCA conditions.
 - C. Uncertainty means a total absence of data.
 - D. Volatility is rapid, unpredictable change while ambiguity is unclear meaning with no precedents.
27. Which statement about learner types is NOT correct? (1)
- A. KNN is a typical eager learner with very fast predictions.
 - B. Eager learners generalize during training.
 - C. Lazy learners store data and generalize at query time.
 - D. Eager learners are memory-efficient at inference time.
28. Which statement about data bias is NOT correct? (1)
- A. Automation bias is over-trusting automated outputs even when wrong.
 - B. Historic bias embeds old societal inequities in the data.
 - C. Anonymized data carries zero risk of re-identification.
 - D. Algorithmic bias reproduces and amplifies biases in the training data.
29. In the Viable System Model, early warning is provided mainly by... (1)
- A. ignoring feedback.
 - B. a star schema.
 - C. deleting historical data.
 - D. anti-oscillatory feedback loops that detect and dampen instability early.

30. Why is domain knowledge essential in Data Science?

(1)

- A. It is needed to interpret the problem, the data and the results correctly.
- B. It only matters for visualization.
- C. It is not essential.
- D. It replaces statistics entirely.